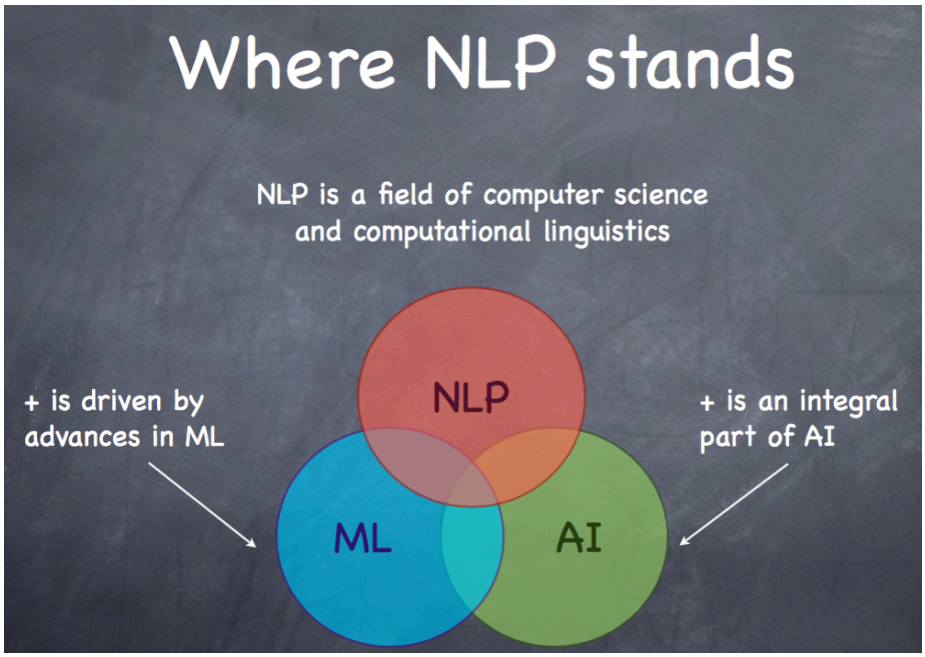
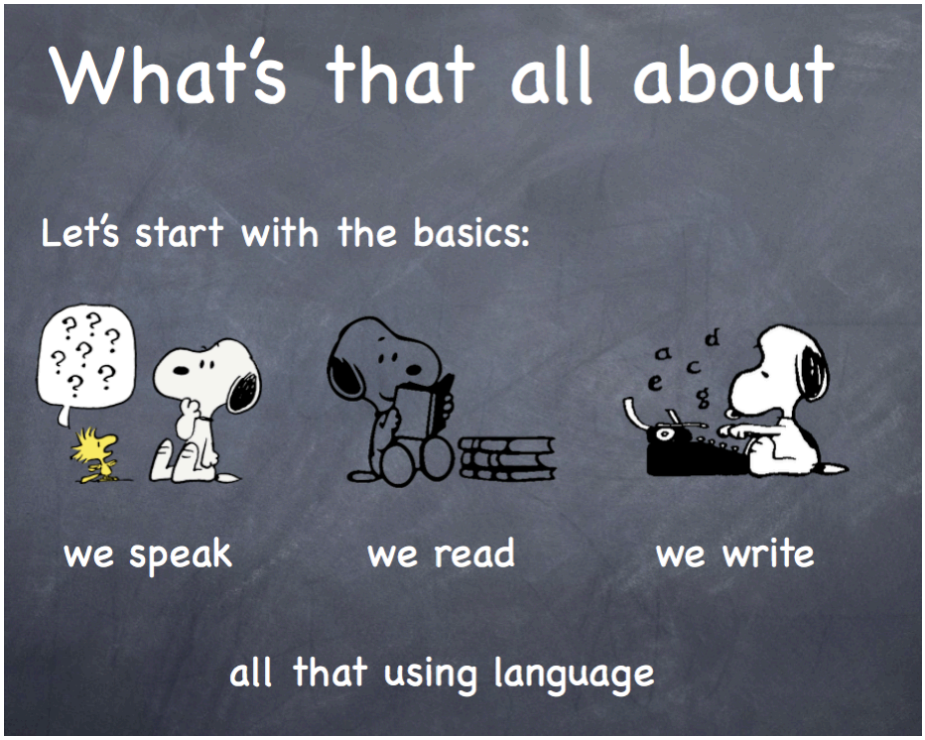
# Introduction to Natural Language processing

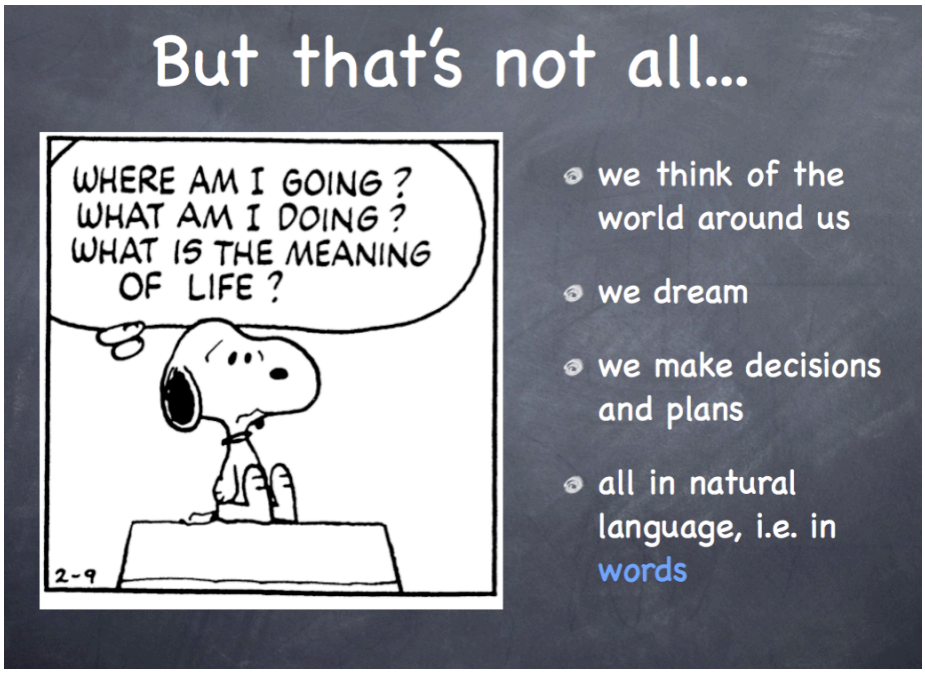
Name: Nawaf Mohammed Alghamdi ID: 2240007352

**Natural language processing (NLP):** Subfield of linguistics, computer science, and information engineering concerned with the interactions between computers and human (natural) languages, in particular how to program computers to process and analyze large amounts of natural language data.



**What is it all about?**






## What are the raw materials for NLP revolution?

**1. Data:** There is a huge amount of human knowledge that has been digitized.

**2. Compute:** Our personal computers are very powerful, and it has never been easier to rent more compute power.

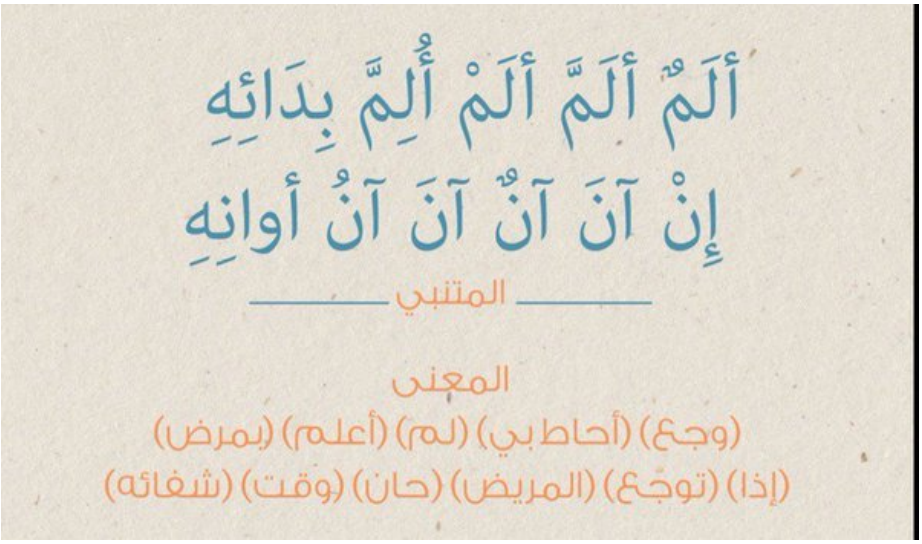
**3. Algorithms:** We use a combination of Computer Science, Machine Learning, Deep Learning, and linguist techniques.

## Why NLP is hard?

It's the nature of the human language that makes NLP difficult.

**Some examples:**

**1. Ambiguity:** One word/sentence can have different meanings.




**2. Spelling Errors:** There are some words that are usually mis-spelled by people.

Spelling-errors.png

**3. Phonetics and Phonology**: Different words that have the same pronunciation.

- there / their
- plane / plain
- ice cream / I scream

**4. Order of Context:** ”Dog bites man” and ”Man bites dog” have vastly different meanings

**In ML terms, why is NLP hard?**

1. **Input:** Large, discrete, open-ended input state spaces
2. **Output:** Variable length output spaces
3. **Algorithms:** Recycled from other fields

## Top Used Text File Types

- CSV
- Text Files
- JSON
- SQL
- HTML
- PDF

- DOCX

## NLP Libraries

There are many popular NLP libraries that we will cover in the next labs. In this lab, we will get introduced to NLTK.

### NLTK

Natural Language Toolkit (nltk) is a Python package for NLP https://www.nltk.org/

**What are Corpus / Corpora?** a collection of written texts, especially the entire works of a particular author or a body of writing on a particular subject.

```
!pip install nltk
```

```
Requirement already satisfied: nltk in /Users/nawaf/venv/lib/python3.11/site-
packages (3.9.4)
Requirement already satisfied: click in /Users/nawaf/venv/lib/python3.11/site-
packages (from nltk) (8.3.3)
Requirement already satisfied: joblib in /Users/nawaf/venv/lib/python3.11/site-
packages (from nltk) (1.5.3)
Requirement already satisfied: regex>=2021.8.3 in /Users/nawaf/venv/lib/python3.11
packages (from nltk) (2026.4.4)
Requirement already satisfied: tqdm in /Users/nawaf/venv/lib/python3.11/site-
packages (from nltk) (4.67.3)


[notice] A new release of pip is available: 24.0 -> 26.2.1
[notice] To update, run: pip install --upgrade pip
```

```
import nltk
```

### Arabic Libraries (Extra)

Camel-Tools: https://camel-tools.readthedocs.io/en/latest/

PyArabic: https://pyarabic.readthedocs.io/ar/latest/

Farasa: https://farasa.qcri.org/

## NLP Applications

### 1. Information Extraction

The automated retrieval of specific information related to a selected topic from a body or bodies of text.

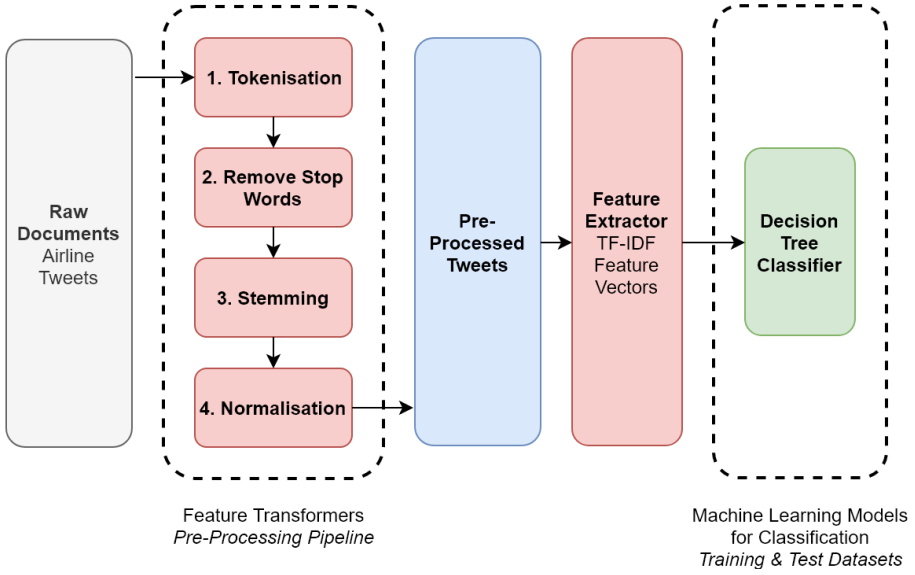
**2. Machine Translation**

The process of using artificial intelligence to automatically translate text from one language to another without human involvement.

**3. Sentiment Analysis**

Identifying sentiments and opinions stated in a text.

## Example of an NLP Pipeline




**Prepared by: Aisha Alansari and Mohammed Al-Hashim**

## Learning Outcomes

```
!pip install nltk
```

```
Requirement already satisfied: nltk in /Users/nawaf/venv/lib/python3.11/site-
packages (3.9.4)
Requirement already satisfied: click in /Users/nawaf/venv/lib/python3.11/site-
packages (from nltk) (8.3.3)
Requirement already satisfied: joblib in /Users/nawaf/venv/lib/python3.11/site-
packages (from nltk) (1.5.3)
Requirement already satisfied: regex>=2021.8.3 in /Users/nawaf/venv/lib/python3.11
packages (from nltk) (2026.4.4)
```

```
Requirement already satisfied: tqdm in /Users/nawaf/venv/lib/python3.11/site-
packages (from nltk) (4.67.3)


[notice] A new release of pip is available: 24.0 -> 26.2.1
[notice] To update, run: pip install --upgrade pip
```

```
import nltk
```